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MATHEMATICS
Paper 1
APRIL, 2026
2 ¼ hours

WAIRAKA MODERN SECONDARY SCHOOL, JINJA

S.4 END OF TERM 1 ASSESSMENT, 2026

Uganda Certificate of Education

MATHEMATICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- ✓ *This paper consists of* **two** *sections;* **A** *and* **B**. *It has* **six** *examination items.*
- ✓ *Section* **A** *has two compulsory items.*
- ✓ *Section* **B** *has two parts;* **I** *and* **II**. *Answer* **one** *item from each part.*
- ✓ *Answer* **four** *examination items in all.*
- ✓ *Any additional item (s) answered will* **not** *be scored.*
- ✓ **All** *answers* **must** *be written in the Answer booklet (s) provided.*
- ✓ *Graph paper is provided.*
- ✓ *Silent, non – programmable scientific calculators and mathematical tables with a list of formulae may be used.*

SECTION A:

Answer **all** *items in this section.*

Item 1.

Joseph is the District Inspector of schools in one of the districts in Eastern region found that his casual workers use **one third** of his farm for bananas, **one quarter** for coffee and **two fifth** of the remainder for mixed farming. He still has some six acres of unused land. A certain school has a student population of **1200** students. On a particular day, joseph invited the entire of $\mathbf{1/5}$ of the boys and $\mathbf{1/4}$ of the girls to go to TRINITY Hall for a sport's meeting and **936** students were left behind. Joseph has a house valued at **45 million** shillings. Its value increased by **25%** after the first year but in the second year, the value of the house depreciated by **10%.**

Task:

(a) Find the size of his farm and clearly illustrate it on a diagram.

(b) Find how many more boys than girls attended the meeting.

(c) Find the value of his house at the end of the second (2nd) year.

Item 2.

A certain school is to transport its S. 4 students for fieldwork in Kasenyi. All the **400** students are to be transported using either coasters or buses. Each coaster can carry **40** people while each bus can carry **80** people. The transport department of the school has only **8** drivers on duty and up to four coasters. If the cost of hiring a coaster is Shs **150,000** and that of hiring a bus is **Shs 300,000**. While in Kasenyi their geography teacher visited a certain shop from which he found that **three** shirts and **two** trousers cost **Shs 105,000** at that shop. **Two** shirts and **five** trousers cost **Shs. 180,000** at the same shop.

Task:

(a) (i) Write down the inequalities representing the above information.

(ii) Represent the inequalities on a graph paper.

(iii) Find the possible number of coasters and buses that can be used and hence determine the minimum cost.

(b) Find the cost of;

(i) each shirt and each trouser.

(ii) three items of each type at the shop.

SECTION B

This Section has two Parts; **I** and **II**

Part I

Answer o**ne** item from this part

Item 3.

There are very few teachers who have three teaching subjects. A survey was done in a certain school and it was found that the school has a teaching staff of **22** teachers. The number of teachers who teach Mathematics, Physics and Chemistry is 8 in each department. **Three** teach both Mathematics and Physics and **one** teaches Mathematics and Chemistry. No teacher teaches all the three subjects. The number of teachers who teach Physics and Chemistry is equal to that of those who teach Physics and mathematics only. In the school staffroom there are two similar cylindrical cans that have different heights. One **50 cm** and the other **75cm**. If the surface area of the larger can is **12,667cm^2** and has a radius **21cm**.

Task:

(a) Find the number of teachers who teach none of the three subjects.

(b) Find the probability that a teacher picked at random teaches only one subject.

(c) Find the surface area of the smaller can.

Item 4.

Three schools want to participate in the National Schools Football Sports Gala to be held in Lyantonde district playground. Unfortunately, none of the schools has a school bus and they want to hire a bus for the one day for the activity. The bus charges **Shs. 15000** per km moved. The three schools through their sports teachers agreed to share the cost of the bus equally amongst themselves. One day they hired a bus from a certain school and they set off at **5:45am** from Kampala and the bus increased the speed gradually to $\mathbf{90}\boldsymbol{km/hr}$ reaching Mpigi at $\mathbf{6{:}45}\boldsymbol{am}$. From there the bus driver maintained this same speed for $\mathbf{1\frac{1}{4}}$ hours reaching Masaka. From Masaka the he reduced slowly in speed reaching Lyantonde at $\mathbf{9{:}30}\boldsymbol{am}$. The games started at 10:00am sharp and each team played six games. School A won **3** games, drew **2** and lost **1** game. School B won **4** games and lost **2** games. School C won 2 games and drew **4** games. The organizers award **three** points for a win, **one** point for a draw and **no** point for a loss. They declared these schools the first three schools in order of their points they obtained from the games. They were to receive the price of money of sixteen million five hundred thousand shillings.

Task:

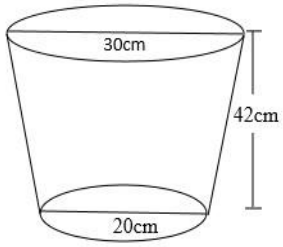
(a) Find how much each school paid for the bus.

(b) Decide the cash prize for each school

Part II
Answer one item from this part

Item 5.

A bucket is in shape of a frustum with an open end of diameter **30**cm and a bottom diameter of **20**cm. the bucket which is 42cm deep is used to fill an empty cylindrical tank of diameter **1.8m** and Height **1.2m**



Taking $\pi = 3.142$.

Three hundred and sixty liters of a homogeneous paint is made by mixing three paints A, B and C. The ratio by amount of point **A** to point **B** is 3:2 and that of **B** to **C** is **1:2**.
Paint **A** costs **Shs 1800** per liter paint **B** costs **Shs 2400** per liter and paint **C Shs 1,275** per liter.

Task:

(a) (i) Determine the capacity of the bucket in liters correct to 3 decimal places.
(ii) The capacity of the tank in liters correct to 2 decimal places.
(iii) The number of buckets that must be drawn to fill the tank.

(b) (i) The amount of each paint in the mixture
(ii) The amount of money needed to make 360 liters of the mixture
(iii) The percentage profit made by selling the mixture at Shs 2,210 per liter.

Item 6.

Akiso bought a car in January 2017 from his friend at **Shs 12,500,000**. If the car depreciates at a rate of 10% per annum. He left Germany for Uganda through Switzerland. While in Switzerland he bought a watch worth 54 Deutsche Marks (Germany currency). Akiso is a secondary school teacher and as a requirement by the government pays PAYE every month according to the tax structure below.

Income (Shs) per month	Tax rate(%
01-50,000	5
50,001 - 100,000	9
100,001 - 180,000	15
180,001 - 300,000	18
300,001 - 400,000	23
400,001 - 500,000	30
Above 500,000	35

He earns **Shs. 760,000** and his allowances include
Marriage allowance - Shs 50,000 per month
Water and electricity - Shs 60,000 per month
Housing allowance - Shs 150,000 per month
Medical allowance - Shs 300,000 per annum
Transport allowance - Shs. 3,000 per day
Paying for insurance and relief - Shs 180,000 per annum Family allowance for only **three** children.
For children in the age bracket **0** to **10** years, **Shs 212,000** per child, Between 10-15 years Shs **9000** per child **15** years and above **Shs 5000**. Given that he has **five** children, two of whom are aged between **0** and **10**, the other two aged between **10** and **15** while the other 18 years.

Support material

Take a month has 30 days
1 Swiss Franc = 1.28 Deutsche Marks
1 Swiss Franc = 1,350 Ugandan Shillings

Task:

(a) Calculate the value of Akiso's car by January 2020.
(b) Find the value of the watch in;
(i) Swiss Francs
(ii) Ugandan Shillings.
(c) Determine his;
(i) taxable income
(ii) Net-income
(d) Determine the percentage of his gross income that goes to tax.

WINNERS DO NOT QUIT

THE END